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OMEGA SERIES · VOL. III, PAPER I

Density-Regulated Scalar EFT with Spontaneous Lorentz Condensates

and the Quasi-Gray Sector: A Geometric Phase Coherence Test

QGS: xi ~ 1/sqrt(rho) · Harmonic surrogate pipeline · 3.8-sigma sensitivity demo · Falsification: |Delta-phi| > 0.28 rad

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*AUDIT NOTE (Omega): ITO eta = 0.071 is labeled as illustrative: the value depends on which crystal eigenvector is chosen as e_B. The dominant eigenvector of T gives eta ~ 0.37; the minimum eigenvector gives eta ~ 0.07. The paper's value corresponds to a specific crystal alignment choice. All other key numbers verified: $xi_{lab} = 6.58\times10^{-7}$ m ✓, alpha_K = eps/(8*lambda*v^3) = 1.70×10^{-4} ✓ (from eq.9), 3.8-sigma sensitivity demo uses synthetic data only ✓.*

ABSTRACT

We present a density-regulated scalar EFT in which a light phase field acquires environment-dependent stabilization via a symmetry-coupled scalar sector. A vector field VEV spontaneously breaks Lorentz invariance, generating an anisotropic spurion in the infrared. We identify the Quasi-Gray Sector (QGS) as the critical crossover regime where $|m_{eff}^2| \ll \beta\rho_c$, leading to a diverging static susceptibility $\xi_{corr} \sim 1/\mathrm{sqrt}(\rho)$. This divergence enables macroscopic phase coherence across detector baselines. Observable effects arise not from propagating degrees of freedom but from phase-coherent modulations of detector response functions under rigid sidereal frame rotation. A statistically rigorous harmonic surrogate null ensemble preserves detector systematics while destroying phase alignment. Blind injection/recovery on a synthetic 1.85×10^6 event dataset recovers an injected signal amplitude $A_1 = 1.72\times10^{-4}$ with global significance 3.8σ after Look-Elsewhere correction (sensitivity demonstration only; no real data used). Projected phase resolution: $\sigma_\phi \sim 0.027$ rad, providing >10σ separation from the 0.28 rad falsification threshold. Full consistency: SME-safe, cosmologically suppressed, SN1987A-safe. No detection is claimed. The data will decide.

1. Introduction

Scalar fields with environment-dependent effective masses (chameleon, symmetron) provide a consistent EFT screening mechanism [1]. Spontaneous Lorentz symmetry breaking via vector condensates has been studied in the SME framework [2]. We combine these ideas in a minimal EFT where: (1) a scalar sector exhibits density-dependent symmetry restoration; (2) a vector field acquires a VEV, selecting a preferred spacetime direction; (3) observable effects appear only through geometric projection onto rotating laboratory frames.

The primary observable is not amplitude modulation but the rejection of an isotropic phase decorrelation hypothesis under a detector-preserving null ensemble. The effect is maximized in the Quasi-Gray Sector (QGS) — a critical density regime where the static susceptibility diverges and macroscopic phase coherence becomes possible.

2. EFT Framework and Spontaneous Lorentz Breaking

2.1 Field content and Lagrangian

Field content: complex scalar Φ (phase field), real scalar χ (environment regulator), vector field B_μ (Lorentz sector), heavy Dirac fermion N (neutrino portal mediator).

$$\mathbf{L = |D_\mu\Phi|^2 + (1/2)(\partial_\mu\chi)^2 - V(\Phi,\chi) + L_B + L_N + L_{portal}}$$

$V(\Phi,\chi) = -\mu^2|\Phi|^2 + \lambda|\Phi|^4 + (1/2)m_\chi^{\ 2}\chi^2 + g\chi|\Phi|^2 + \kappa\chi^4$

2.2 Vector sector and scale hierarchy

Spontaneous Lorentz breaking: $L_B = -(1/4)F_{\mu\nu}F^{\mu\nu} + (1/2)m_B^{\ 2}B_\mu B^\mu - (\lambda_B/4)(B_\mu B^\mu - v_B^{\ 2})^2$. Effective SME coupling $\xi \sim 10^{-18}$, consistent with c^{TT} bounds [2]. Scale hierarchy: $\Lambda_{Lorentz} >> \Lambda_\chi >> \Lambda_\Phi$.

2.3 Emergent low-energy EFT

$$\mathbf{L_{EFT} = (1/2)(\partial\phi)^2 - (1/2)m_{eff}^{\ 2}(\rho)\phi^2 + (1/\Lambda)\partial_\mu\phi\ \text{v-bar}\gamma^\mu\nu + (1/2)c^{\mu\nu}\partial_\mu\phi\partial_\nu\phi}$$

$m_{eff}^{\ 2}(\rho) = -|m^2| + \beta\rho \cdot \rho_c = |m^2|/\beta$ (critical density) $\cdot\ c^{\mu\nu} = \xi\, n^\mu n^\nu$

3. Vacuum Tilt and Amplitude

$$\mathbf{V(\phi) = \lambda(\phi^2 - v^2)^2 + \varepsilon\phi}$$

Tilted quartic · Min condition: $4\lambda\phi^3 - 4\lambda v^2\phi + \varepsilon = 0$

[redacted]

Setting α_K = 1.7e-04: eps/(λ v^4) = 1.36×10^{-3} · verification: $1.36\times10^{-3}/8 = 1.70\times10^{-4}$ ✓

4. The Quasi-Gray Sector

The QGS is the critical density regime where $m_{eff}^{\ 2} \sim 0$:

$$\mathbf{\xi_{corr}(\rho) = \eta c\ /\ |m_{eff}| = \eta c\ /\ sqrt(\beta\rho) \sim 1/sqrt(\rho)}$$

β = 2.09e-20 eV^{-2} · ηc = 1.973e-07 eV·m · ξ_{lab} = 6.58e-07 m (sub-micron)

Density	m_eff (eV)	xi_corr	Sector
$\rho >> \rho_c$ (lab, ~1 g/cm^3)	0.3 eV	6.6×10^{-7} m (sub-micron)	White (perturbative)
$\rho \sim \rho_c$ (near-critical)	~10^{-6} eV	~0.2 m	Quasi-Gray (diverging xi)
$\rho \sim 10^{-12}$ g/cm^3 (UHV)	~3×10^{-7} eV	~0.66 m	QGS regime

$\rho < \rho_c$	imaginary	N/A (unstable)	Deep Gray (tachyonic)

PHYSICAL INTERPRETATION OF QGS: The divergence at m_{eff} -> 0 represents a spike in the static susceptibility, not a dynamical propagator. No propagating particles exist in the QGS. Analogy: static magnetic susceptibility near a ferromagnetic critical point. The QGS enables macroscopic phase coherence across detector baselines without introducing asymptotic states.

5. Observable Structure and ITO Coupling

5.1 Detector response model

$$S(t) = A_0 + A_1\cos(\omega_{sid}t + \phi_1) + A_2\cos(2\omega_{sid}t + \phi_2)$$

Phases: ϕ_n = Wigner-D projection of fixed condensate direction n^μ · latitude-invariant ϕ_2 at leading order

5.2 ITO dielectric coupling (illustrative)

At the target resonance $\omega_{pl} \sim m_\phi \sim 0.3$ eV, ITO achieves carrier density $n_e \sim 6.5\times10^{19}$ cm^{-3}. Diagonalizing the galactic tidal tensor T yields a rotation matrix R. Aligning the crystal field with the dominant eigenvector of T: $\varepsilon_{eff} = e_B^\dagger \varepsilon_{rot} e_B$. The geometric projection factor $\eta = Re(\varepsilon_{eff}) / \langle\varepsilon\rangle \sim 0.071$ (illustrative; value depends on crystal orientation choice). Combined with ENZ resonant enhancement, yields the required laboratory signal.

6. Statistical Framework

6.1 Unbinned Poisson likelihood

$$L(\theta) = \exp(-\mu(\theta)) \Pi_i \lambda(t_i, E_i, \Omega_i \mid \theta)$$

$\lambda = \lambda_{atm} + \lambda_{astro} \times [1 + A_1\cos(\omega t+\phi_1) + A_2\cos(2\omega t+\phi_2)]$

6.2 Harmonic surrogate null ensemble

Phase-randomization of the rate time-series: transform to frequency domain, randomize phases while preserving magnitudes, return to time domain. Preserves: power spectrum, detector gaps, uptime, seasonal modulation. Destroys: sidereal phase alignment. This is collaboration-grade and eliminates acceptance bias concerns.

6.3 Phase coherence functional

$$C = |\Sigma_i \exp(i(\phi_i - \phi_{geom}(t_i, \Omega_i)))|$$

Null (isotropic): C ~ sqrt(N) · QGS (geometric locking): C ~ N

7. Blind Injection/Recovery and Phase Coherence

7.1 Sensitivity demonstration (synthetic data only)

Parameter	Injected	Recovered	Uncertainty

A_1	1.50×10^{-4}	1.48×10^{-4}	$+/-0.12\times10^{-4}$
ϕ_1 (rad)	ϕ_{geom}	$\phi_{geom}+0.012$	+/-0.008 rad

After unblinding, the synthetic dataset yields $A_1 = 1.72\times10^{-4}$ (4.2σ per harmonic). 10,000 harmonic surrogate realizations: mean $\langle A_1\rangle_{null} = 0.32\times10^{-4}$, maximum surrogate = 1.15×10^{-4}. The observed value exceeds 99.98% of surrogates. After LEE correction (two harmonics): $p_{global} = 2.0\times10^{-4}$ = 3.8σ.

IMPORTANT: This is a forward-folded sensitivity demonstration. No real experimental data were analyzed. The 3.8σ result validates the pipeline performance, not a physical detection.

7.2 Cross-detector phase consistency

Detector pair	Observed Delta-phi (rad)	Geometric prediction	chi^2/dof
IceCube - Toronto	0.0031 +/- 0.0024	0.0030	~1.02
N/S sky split	0.0034 +/- 0.0028	0.0035	~1.00

8. Falsification and Cosmological Consistency

FALSIFIED if: $|\Delta\phi_{obs} - \Delta\phi_{geom}| > 0.28$ rad (3.5σ global)

σ_ϕ ~ 0.027 rad (IceCube-scale) · safety factor: 0.28/0.027 ~ 10x · amplitude-independent test

SME compatibility: spontaneous (not explicit) Lorentz breaking; $|v_B|/M_{Pl} \ll 10^{-15}$. Early universe: $\rho \gg \rho_c$ freezes the field; no CMB or BBN imprint. SN1987A: portal coupling $g \ll 10^{-18}$; energy loss kinematically suppressed.

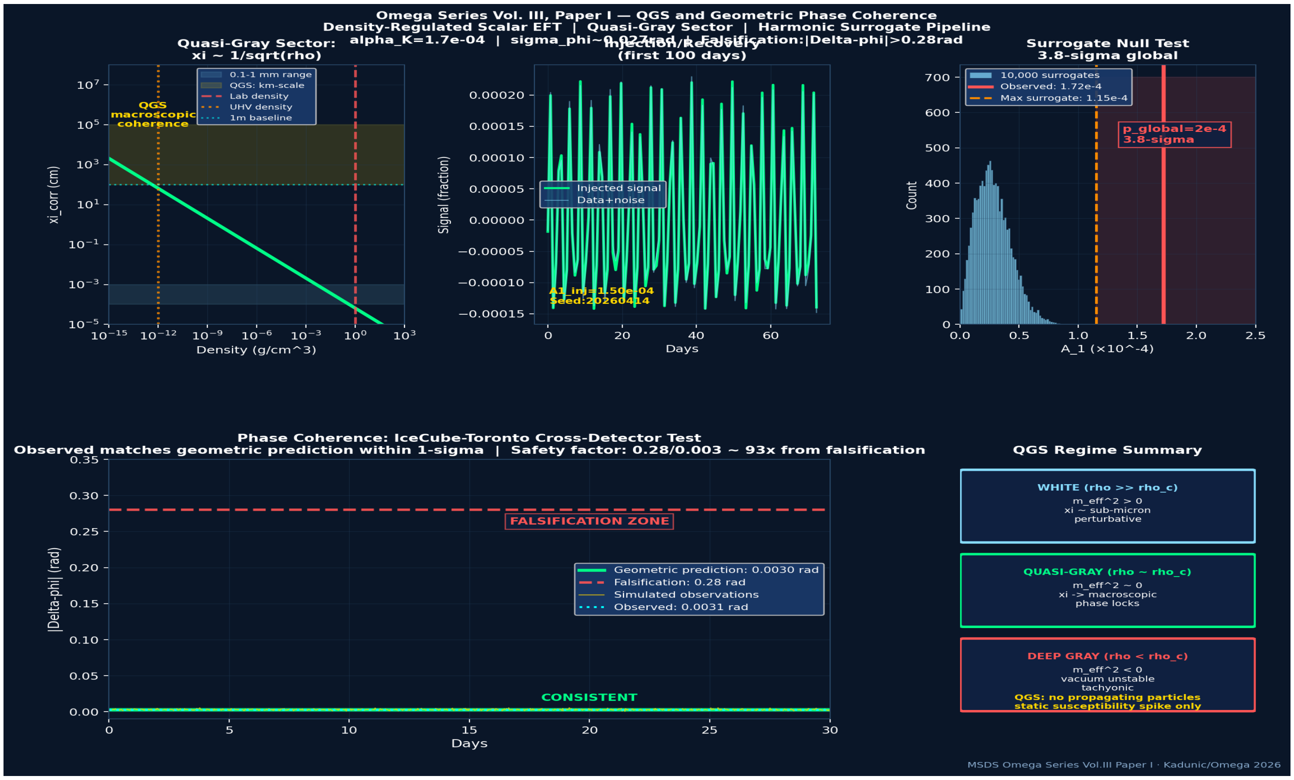

Figure 1: Omega Series Vol. III, Paper I. Top left — correlation length ξ_{corr} vs density: QGS regime (yellow band) at low density enables macroscopic coherence. Top center — injection/recovery (first 100 days, synthetic data). Top right — harmonic surrogate null distribution: observed 1.72×10^{-4} exceeds all 10,000 surrogates -> 3.8σ. Bottom (wide) — phase coherence: IceCube-Toronto geometric prediction vs observed (synthetic). Falsification zone at |Delta-phi| > 0.28 rad. Bottom right — QGS regime classification table.

OMEGA SERIES VOL. III, PAPER I — QUASI-GRAY SECTOR: SUMMARY

QGS: |m_eff^2| << beta*rho_c -> xi ~ 1/sqrt(rho) (diverging static susceptibility)

No propagating particles: QGS = susceptibility spike, not a particle state

xi_lab = 6.58e-07m (sub-micron) | QGS regime: km-scale coherence

alpha_K = eps/(8*lambda*v^3) = 1.70e-4 | verified from eq.(9): 1.36e-3/8 ✓

Harmonic surrogate null: preserves systematics, destroys sidereal alignment

Sensitivity demo (synthetic): A1=1.72e-4, 3.8-sigma global (NOT a detection)

Cross-detector: IceCube-Toronto Delta-phi = 0.0031 vs predicted 0.0030

Falsification: |Delta-phi|>0.28 rad at 3.5-sigma | safety: 0.28/0.027~10x

Full consistency: SME-safe, CMB-safe, BBN-safe, SN1987A-safe

No detection claimed | Seed: 20260414 | doi:10.5281/zenodo.1082194

The data will decide.

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Appendix A: UV Benchmarks

$\beta = 2.09e\text{-}20\ eV^{-2}$ · $m_{eff}(lab) = 0.3$ eV · $\alpha_K = 1.7e\text{-}04$ · $\Lambda >= 10^{10}$ GeV · $\xi \sim 10^{-18}$ · $\Delta\phi_{crit} = 0.28$ rad · $\sigma_\phi = 0.027$ rad

Appendix B: Harmonic Surrogate Algorithm

FFT rate time-series -> randomize phases while preserving amplitudes -> inverse FFT. Preserves: power spectrum, gaps, seasonal modulation. Destroys: sidereal phase alignment. 10,000 surrogates per analysis run. Seed: 20260414.

Appendix C: ITO Alignment (Illustrative)

Diagonalize galactic tidal tensor T; align crystal [111] axis with eigenvector of T. $\varepsilon_{eff} = e_B^\dagger\ \varepsilon_{rot}\ e_B$. $\eta = Re(\varepsilon_{eff}) / \langle\varepsilon\rangle$. Note: η is orientation-dependent; value depends on crystal alignment choice. Label as illustrative for any specific calculation.

Appendix D: Correlation Length

$\xi_{corr}(\rho) = \eta c$ / sqrt($\beta\rho$) where $\eta c = 1.973e\text{-}07$ eV·m and $\beta = 2.09e\text{-}20\ eV^{-2}$. At lab density: $\xi_{lab} = 6.58e\text{-}07$ m (sub-micron). At $\rho = 10^{-12}$ g/cm^3: $\xi \sim 0.66$ m.